

The Existence
Theorem

Taylor Polynomials

Taylor's Theorem

The Uniqueness
Theorem

The exp Function

The Trigonometric
Functions

The number π

The Hyperbolic
Functions

Euler's Formula

The Polar
Representation of $\mathbb{C}$

De Moivre's
Formula

The n th Roots of
Unity

Elementary Functions

Complex Analysis

Morales Carmona Ronny

Autumn 2026

The Existence
Theorem

Taylor Polynomials

Taylor's Theorem

The Uniqueness
Theorem

The exp Function

The Trigonometric
FunctionsThe number π The Hyperbolic
Functions

Euler's Formula

The Polar
Representation of $\mathbb{C}$ De Moivre's
FormulaThe n th Roots of
Unity

Theorem 1.1

There exists a function $E : \mathbb{R} \rightarrow \mathbb{R}$ such that:

- i) $E'(x) = E(x)$ for all $x \in \mathbb{R}$.
- ii) $E(0) = 1$.

The Existence Theorem

Taylor Polynomials

Taylor's Theorem

The Uniqueness Theorem

The exp Function

The Trigonometric Functions

The number π

The Hyperbolic Functions

Euler's Formula

The Polar Representation of $\mathbb{C}$

De Moivre's Formula

The n th Roots of Unity

Corollary 1.2

The function E has a derivative of every order and $E^{(n)}(x) = E(x)$ for all $n \in \mathbb{N}, x \in \mathbb{R}$.

Definition

If a function f has an n th derivative at a point $x_0 \in \mathbb{R}$, it is not difficult to construct an n th degree polynomial P_n such that $P_n(x_0) = f(x_0)$ and $P_n^{(k)}(x_0) = f^{(k)}(x_0)$ for $k = 1, 2, \dots, n$. In fact, the polynomial

$$P_n(x) := f(x_0) + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \dots + \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n$$

has the property that it and its derivatives up to order n agree with the function f and its derivatives up to order n at the specified point x_0 . This polynomial P_n is called the **n th Taylor polynomial for f at x_0** . It is natural to expect this polynomial to provide a reasonable approximation to f for points near x_0 , but to gauge the quality of the approximation, it is necessary to have information concerning the remainder $R_n := f - P_n$. The following fundamental result provides such information.

Theorem 1.3 (Taylor's Theorem)

Let $n \in \mathbb{N}$, let $I := [a, b]$, and let $f : I \rightarrow \mathbb{R}$ be such that f and its derivatives $f', f'', \dots, f^{(n)}$ are continuous on I and that $f^{(n+1)}$ exists on (a, b) . If $x_0 \in I$, then for any $x \in I$ there exists a point c between x and x_0 such that

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + \frac{f^{(n+1)}(c)}{(n+1)!} (x - x_0)^{n+1}.$$

The formula

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!} (x - x_0)^{n+1}$$

is referred to as the **Lagrange form** (or the **derivative form**) of the remainder.

[The Existence
Theorem](#)[Taylor Polynomials](#)[Taylor's Theorem](#)[The Uniqueness
Theorem](#)[The exp Function](#)[The Trigonometric
Functions](#)[The number \$\pi\$](#) [The Hyperbolic
Functions](#)[Euler's Formula](#)[The Polar
Representation of \$\mathbb{C}\$](#) [De Moivre's
Formula](#)[The \$n\$ th Roots of
Unity](#)

Lemma 1.4

The unique function $F : \mathbb{R} \rightarrow \mathbb{R}$ satisfying $F'(x) = F(x)$ and $F(0) = 1$ is the exponential function given by $F(x) = e^x$ for all $x \in \mathbb{R}$.

Theorem 1.5

The function $E : \mathbb{R} \rightarrow \mathbb{R}$ that satisfies (i) and (ii) of the previous theorem is unique.

Theorem 1.6

The exponential function satisfies the following properties:

- iii) $E(x) \neq 0$ for all $x \in \mathbb{R}$;
- iv) $E(x + y) = E(x)E(y)$ for all $x, y \in \mathbb{R}$;
- v) $E(rx) = (E(x))^r$ for all $x \in \mathbb{R}, r \in \mathbb{Q}$.

Definition

The unique function $E : \mathbb{R} \rightarrow \mathbb{R}$, such that $E'(x) = E(x)$ for all $x \in \mathbb{R}$ and $E(0) = 1$, is called the **exponential function**. The number $e := E(1)$ is called **Euler's number**. We will frequently write

$$\exp x := E(x) \quad \text{or} \quad e^x := E(x) \quad \text{for} \quad x \in \mathbb{R}.$$

Definition

The function defined by the series

$$E(z) := \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

*converges uniformly on $\mathbb{C}$, absolutely for all $z \in \mathbb{C}$ and is called the **complex exponential function**[†]. We will frequently write*

$$\exp z := E(z) \quad \text{or} \quad e^z := E(z) \quad \text{for} \quad z \in \mathbb{C}.$$

[†]Since we will be working mostly on the complex realm, we may refer to the complex exponential function simply as the exponential function.

The Existence
Theorem

Taylor Polynomials

Taylor's Theorem

The Uniqueness
Theorem

The exp Function

The Trigonometric
Functions

The number π

The Hyperbolic
Functions

Euler's Formula

The Polar
Representation of $\mathbb{C}$

De Moivre's
Formula

The n th Roots of
Unity

Theorem 1.7

The complex exponential function is unique for each $z \in \mathbb{C}$ and satisfies the extension of properties (i), (ii), (iii), and (iv).

The Existence
Theorem

Taylor Polynomials

Taylor's Theorem

The Uniqueness
Theorem**The exp Function**The Trigonometric
FunctionsThe number π The Hyperbolic
Functions

Euler's Formula

The Polar
Representation of $\mathbb{C}$ De Moivre's
FormulaThe n th Roots of
Unity**Example 1.8**

Show that the modulus of $\exp z$ is $\exp \Re(z)$. That is, the modulus of the exponential function is only determined by the real part of the argument.

The Existence
Theorem

Taylor Polynomials

Taylor's Theorem

The Uniqueness
Theorem

The exp Function

**The Trigonometric
Functions**The number π The Hyperbolic
Functions

Euler's Formula

The Polar
Representation of $\mathbb{C}$ De Moivre's
FormulaThe n th Roots of
Unity

Theorem 1.9

There exist functions $C : \mathbb{R} \rightarrow \mathbb{R}$ and $S : \mathbb{R} \rightarrow \mathbb{R}$ such that

- i) $C''(x) = -C(x)$ and $S''(x) = -S(x)$ for all $x \in \mathbb{R}$.*
- ii) $C(0) = 1$, $C'(0) = 0$, and $S(0) = 0$, $S'(0) = 1$.*

Corollary 1.10

If C, S are the functions from the previous theorem, then

iii) $C'(x) = -S(x)$ and $S'(x) = C(x)$ for $x \in \mathbb{R}$.

Moreover, these functions have derivatives of all orders.

The Existence
Theorem

Taylor Polynomials

Taylor's Theorem

The Uniqueness
Theorem

The exp Function

**The Trigonometric
Functions**The number π The Hyperbolic
Functions

Euler's Formula

The Polar
Representation of $\mathbb{C}$ De Moivre's
FormulaThe n th Roots of
Unity**Corollary 1.11**

*The functions C and S satisfy the **Pythagorean Identity**:*

iv) $(C(x))^2 + (S(x))^2 = 1$ for $x \in \mathbb{R}$.

The Existence
Theorem

Taylor Polynomials

Taylor's Theorem

The Uniqueness
Theorem

The exp Function

**The Trigonometric
Functions**

The number π

The Hyperbolic
Functions

Euler's Formula

The Polar
Representation of $\mathbb{C}$

De Moivre's
Formula

The n th Roots of
Unity

Theorem 1.12

The functions C and S satisfying properties (i) and (ii) are unique.

Definition

The unique functions $C : \mathbb{R} \rightarrow \mathbb{R}$ and $S : \mathbb{R} \rightarrow \mathbb{R}$ such that $C''(x) = -C(x)$ and $S''(x) = -S(x)$ for all $x \in \mathbb{R}$ and $C(0) = 1, C'(0) = 1$, and $S(0) = 0, S'(0) = 1$, are called the **cosine function** and the **sine function**, respectively. We ordinarily write

$$\cos x := C(x) \quad \text{and} \quad \sin x := S(x) \quad \text{for} \quad x \in \mathbb{R}.$$

[The Existence
Theorem](#)[Taylor Polynomials](#)[Taylor's Theorem](#)[The Uniqueness
Theorem](#)[The exp Function](#)[The Trigonometric
Functions](#)[The number \$\pi\$](#) [The Hyperbolic
Functions](#)[Euler's Formula](#)[The Polar
Representation of \$\mathbb{C}\$](#) [De Moivre's
Formula](#)[The \$n\$ th Roots of
Unity](#)

Lemma 1.13

There exists a root ϖ of the cosine function in the interval $(\sqrt{2}, \sqrt{3})$. Moreover $C(x) > 0$ for $x \in [0, \varpi)$. The number 2ϖ is the smallest positive root of S .

The Existence
Theorem

Taylor Polynomials

Taylor's Theorem

The Uniqueness
Theorem

The exp Function

The Trigonometric
Functions

The number π

The Hyperbolic
Functions

Euler's Formula

The Polar
Representation of $\mathbb{C}$

De Moivre's
Formula

The n th Roots of
Unity

Definition

*Let $\pi := 2\varpi$ denote the smallest positive root of S . We will often refer to π as the **number pi***

Definition

The unique functions $c : \mathbb{R} \rightarrow \mathbb{R}$ and $s : \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

i) $c''(x) = c(x)$ and $s''(x) = s(x)$ for all $x \in \mathbb{R}$;

ii) $c(0) = 1, c'(0) = 0$ and $s(0) = 0, s'(0) = 1$;

are called the **hyperbolic cosine** and **hyperbolic sine functions**, respectively.

The hyperbolic cosine and hyperbolic sine functions are commonly denoted by $\cosh$ and $\sinh$, respectively.

Definition

We define the extensions of the sine, hyperbolic sine, cosine and hyperbolic cosine functions by

$$\sin z = \frac{1}{2i} (e^{iz} - e^{-iz}),$$

$$\cos z = \frac{1}{2} (e^{iz} + e^{-iz}),$$

$$\sinh z = \frac{e^z - e^{-z}}{2},$$

$$\cosh z = \frac{e^z + e^{-z}}{2}.$$

The Existence
Theorem

Taylor Polynomials

Taylor's Theorem

The Uniqueness
Theorem

The exp Function

The Trigonometric
Functions

The number π

**The Hyperbolic
Functions**

Euler's Formula

The Polar
Representation of $\mathbb{C}$

De Moivre's
Formula

The n th Roots of
Unity

Example 1.14

Find a power series representation for $\sin z$ and $\cos z$.

Proposition 1.15

The functions $\sin : \mathbb{C} \rightarrow \mathbb{C}$ and $\cos : \mathbb{C} \rightarrow \mathbb{C}$ are entire, unique, and satisfy:

I) $(\sin z)' = \cos z;$

II) $(\cos z)' = -\sin z;$

III) $\sin^2 z + \cos^2 z = 1;$

IV) $\sin(z + w) = \sin z \cos w + \sin w \cos z;$

V) $\cos(z + w) = \cos z \cos w - \sin z \sin w;$

for all $z, w \in \mathbb{C}$.

The Existence
Theorem

Taylor Polynomials

Taylor's Theorem

The Uniqueness
Theorem

The exp Function

The Trigonometric
FunctionsThe number π **The Hyperbolic
Functions**

Euler's Formula

The Polar
Representation of $\mathbb{C}$ De Moivre's
FormulaThe n th Roots of
Unity

Example 1.16

Use the definition of π and properties (III), (IV), (V) to show that:

- $\sin 2\pi = 0$;
- $\cos 2\pi = 1$;
- $\sin \pi/4 = \cos \pi/4 = \frac{1}{\sqrt{2}}$.

[The Existence
Theorem](#)[Taylor Polynomials](#)[Taylor's Theorem](#)[The Uniqueness
Theorem](#)[The exp Function](#)[The Trigonometric
Functions](#)[The number \$\pi\$](#) [The Hyperbolic
Functions](#)[Euler's Formula](#)[The Polar
Representation of \$\mathbb{C}\$](#) [De Moivre's
Formula](#)[The \$n\$ th Roots of
Unity](#)

Example 1.17

Show that the cosine and sine function are even and odd, respectively, in the sense that

VI) $C(-z) = C(z)$ for $z \in \mathbb{C}$;

VII) $S(-z) = -S(z)$ for $z \in \mathbb{C}$.

The Existence
Theorem

Taylor Polynomials

Taylor's Theorem

The Uniqueness
Theorem

The exp Function

The Trigonometric
Functions

The number π

**The Hyperbolic
Functions**

Euler's Formula

The Polar
Representation of $\mathbb{C}$

De Moivre's
Formula

The n th Roots of
Unity

Remark

Unlike the real sine function, the complex sine extension is not bounded in modulus by 1, and the same is true for the complex extension of the cosine function, in fact, their moduli are unbounded.

[The Existence Theorem](#)[Taylor Polynomials](#)[Taylor's Theorem](#)[The Uniqueness Theorem](#)[The exp Function](#)[The Trigonometric Functions](#)[The number \$\pi\$](#) [The Hyperbolic Functions](#)[Euler's Formula](#)[The Polar Representation of \$\mathbb{C}\$](#) [De Moivre's Formula](#)[The \$n\$ th Roots of Unity](#)

Theorem 1.18 (Euler's Formula)

The equation

$$e^{i\theta} = \cos \theta + i \sin \theta$$

*is called **Euler's equation** (or **Euler's formula**) and holds for all $\theta \in \mathbb{R}$.*

[The Existence
Theorem](#)[Taylor Polynomials](#)[Taylor's Theorem](#)[The Uniqueness
Theorem](#)[The exp Function](#)[The Trigonometric
Functions](#)[The number \$\pi\$](#) [The Hyperbolic
Functions](#)[Euler's Formula](#)[The Polar
Representation of \$\mathbb{C}\$](#) [De Moivre's
Formula](#)[The \$n\$ th Roots of
Unity](#)

Example 1.19

*Prove the celebrated **Euler's identity***

$$e^{i\pi} + 1 = 0.$$

The Existence
Theorem

Taylor Polynomials

Taylor's Theorem

The Uniqueness
Theorem

The exp Function

The Trigonometric
Functions

The number π

The Hyperbolic
Functions

Euler's Formula

**The Polar
Representation of $\mathbb{C}$**

De Moivre's
Formula

The n th Roots of
Unity

Theorem 1.20

Given any $z \in \mathbb{C}$ with $z \neq 0$, there exists unique $r > 0$ and $w \in \mathbb{C}$, with $|w| = 1$, such that $z = rw$.

[The Existence
Theorem](#)[Taylor Polynomials](#)[Taylor's Theorem](#)[The Uniqueness
Theorem](#)[The exp Function](#)[The Trigonometric
Functions](#)[The number \$\pi\$](#) [The Hyperbolic
Functions](#)[Euler's Formula](#)[The Polar
Representation of \$\mathbb{C}\$](#) [De Moivre's
Formula](#)[The \$n\$ th Roots of
Unity](#)

Theorem 1.21

Let $w \in \mathbb{C}$ be such that $|w| = 1$. Then there exists a unique value $\theta \in \mathbb{R}$ such that $-\pi < \theta \leq \pi$ and $w = \exp(i\theta)$.

The Existence
Theorem

Taylor Polynomials

Taylor's Theorem

The Uniqueness
Theorem

The exp Function

The Trigonometric
Functions

The number π

The Hyperbolic
Functions

Euler's Formula

**The Polar
Representation of $\mathbb{C}$**

De Moivre's
Formula

The n th Roots of
Unity

Theorem 1.22

$\exp(z) = 1$ if and only if $z = 2k\pi i$ for some $k \in \mathbb{Z}$.

[The Existence
Theorem](#)[Taylor Polynomials](#)[Taylor's Theorem](#)[The Uniqueness
Theorem](#)[The exp Function](#)[The Trigonometric
Functions](#)[The number \$\pi\$](#) [The Hyperbolic
Functions](#)[Euler's Formula](#)[The Polar
Representation of \$\mathbb{C}\$](#) [De Moivre's
Formula](#)[The \$n\$ th Roots of
Unity](#)

Corollary 1.23

Let $z \in \mathbb{C}$, $z \neq 0$. Then there exist a unique $r > 0$, and a unique $\theta \in (-\pi, \pi]$ such that $z = r \exp(i\theta)$. In fact, $r = |z|$.

The Existence
Theorem

Taylor Polynomials

Taylor's Theorem

The Uniqueness
Theorem

The exp Function

The Trigonometric
FunctionsThe number π The Hyperbolic
Functions

Euler's Formula

**The Polar
Representation of $\mathbb{C}$** De Moivre's
FormulaThe n th Roots of
Unity

Definition

If $z \in \mathbb{C}$ and if $z \neq 0$, then the unique value $\theta \in (-\pi, \pi]$ for which $z = r \exp(i\theta)$ is called the **principal argument** of z and is denoted by $\text{Arg } z$. Any other value $\theta' \notin (-\pi, \pi]$ for which $z = r \exp(i\theta')$ is called an **argument** of z and denoted by $\arg z$.

Definition

If a complex number $z \neq 0$ is written in the form

$$z = r \exp(i\theta), \quad r = |z|, \quad \theta = \operatorname{Arg} z,$$

then we say that z is in its **polar representation**.

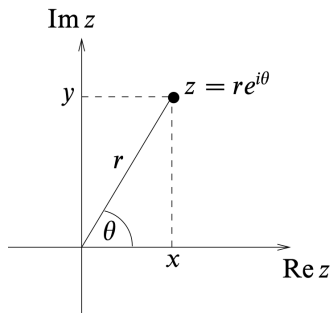


Figure 1: The polar representation of a complex number $z \neq 0$

The Existence
Theorem

Taylor Polynomials

Taylor's Theorem

The Uniqueness
Theorem

The exp Function

The Trigonometric
Functions

The number π

The Hyperbolic
Functions

Euler's Formula

**The Polar
Representation of $\mathbb{C}$**

De Moivre's
Formula

The n th Roots of
Unity

Remark

The relations

$$|r_1 \exp(i\theta_1)r_2 \exp(i\theta_2)| = r_1 r_2 \quad \arg(r_1 \exp(i\theta_1)r_2 \exp(i\theta_2)) = \theta_1 + \theta_2$$

hold true for every pair of complex numbers z_1, z_2 in their polar representation. Thus we identify the multiplication by a complex number $z = r \exp(i\theta)$ as a rotation (counterclockwise by an angle θ) plus an elongation (by r).

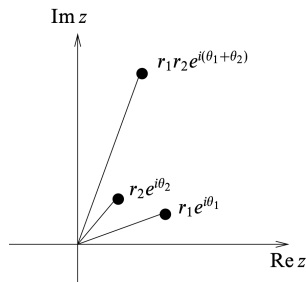


Figure 2: The polar representation of multiplication in $\mathbb{C}$

The Existence
Theorem

Taylor Polynomials

Taylor's Theorem

The Uniqueness
Theorem

The exp Function

The Trigonometric
Functions

The number π

The Hyperbolic
Functions

Euler's Formula

**The Polar
Representation of $\mathbb{C}$**

De Moivre's
Formula

The n th Roots of
Unity

Remark

Similarly, since division can be thought of as the inverse operation of multiplication, we identify division of complex numbers as a “reverse” rotation (now clockwise by an angle θ) plus a “contraction” (by $1/r$).

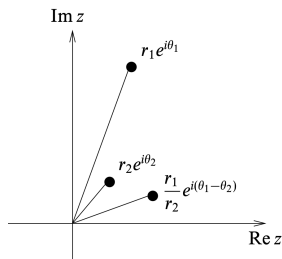


Figure 3: The polar representation of division in $\mathbb{C}$

[The Existence
Theorem](#)[Taylor Polynomials](#)[Taylor's Theorem](#)[The Uniqueness
Theorem](#)[The exp Function](#)[The Trigonometric
Functions](#)[The number \$\pi\$](#) [The Hyperbolic
Functions](#)[Euler's Formula](#)[The Polar
Representation of \$\mathbb{C}\$](#) [**De Moivre's
Formula**](#)[The \$n\$ th Roots of
Unity](#)

Theorem 1.24 (De Moivre's Formula)

Let $z \in \mathbb{C}$, let $r = |z|$ and let $\theta = \arg z$, then

$$z^n = r^n (\cos n\theta + i \sin n\theta)$$

[The Existence
Theorem](#)[Taylor Polynomials](#)[Taylor's Theorem](#)[The Uniqueness
Theorem](#)[The exp Function](#)[The Trigonometric
Functions](#)[The number \$\pi\$](#) [The Hyperbolic
Functions](#)[Euler's Formula](#)[The Polar
Representation of \$\mathbb{C}\$](#) [De Moivre's
Formula](#)[The \$n\$ th Roots of
Unity](#)

Definition

*Each complex number ζ such that $\zeta^n = 1$ is called an **n th root of unity**. Similarly, given $z \in \mathbb{C}$, each complex number w such that $w^n = z$ is called an **n th root of z** .*

The Existence
Theorem

Taylor Polynomials

Taylor's Theorem

The Uniqueness
Theorem

The exp Function

The Trigonometric
FunctionsThe number π The Hyperbolic
Functions

Euler's Formula

The Polar
Representation of $\mathbb{C}$ De Moivre's
Formula**The n th Roots of
Unity**

Theorem 1.25 (The n th Roots of Unity)

Given $n \in \mathbb{N}$, the set of all n th roots of unity is given by

$$\mathcal{R} = \{\zeta^k\}_{k=0}^{n-1}, \quad \zeta := e^{2\pi i/n}.$$

Theorem 1.26

Given $n \in \mathbb{N}$, $z \in \mathbb{C}$, the set $z^{1/n}$ of all n th roots of z is given by

$$z^{1/n} = \{te^{i\theta/n}\zeta^k\}_{k=0}^{n-1},$$

where t is the unique positive real number such that $t^n = |z|$, θ is the principal argument of z and ζ is defined as in the previous theorem.